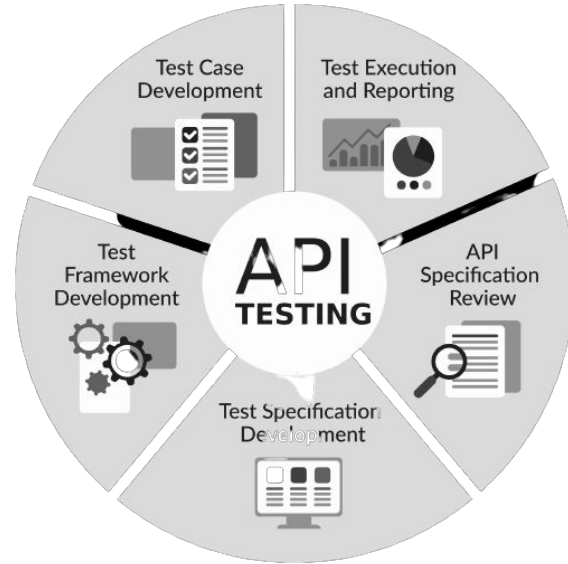


API Testing Mastery



Pramod Dutta
Lead SDET.



API Testing - Get Started!!

Agenda of Session

- Who should learn API Testing
- Why We should Learn API Testing?
- What is API?
- What is API Testing?
- Types of APIs
- SOAP Vs REST
- What To Test in APIs?
- Manual Testing Concepts to Design API Test cases.

**Before We Start.....
Join Private Telegram**

<https://sdet.live/apibatch>

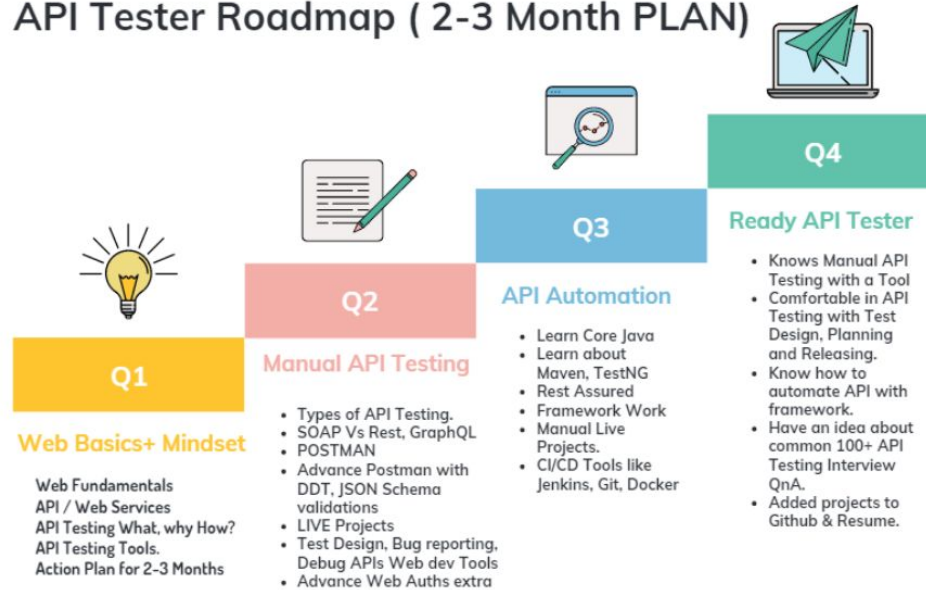
API Tester ROADMAP



Made by Pramod(TheTestingAcademy)

<https://apitesting.co/blueprint>

API Tester Roadmap (2-3 Month PLAN)



<https://apitesting.co/blueprint>



Who should learn API Testing

- If you are Manual Tester? -> API Testing is Must to have
- If you are Automation Tester - API Testing is Must
- Fresher? -> Must have API Testing
- You can switch from Manual to Automation using APIs.
- Non IT want to start to IT Software Testing.

Why we need to Learn API Testing?

- Companies are moving towards Microservices Architecture.
- API Testing is must to have skill nowadays for Freshers, Manual Tester and Automation Testers or Devs.
- Many Companies are especially hiring API Testers.
- API Test are more reliable, easy to create, and stable, less complex and provide more coverage as Business layer level.



JOBs in API Testing

<https://www.linkedin.com/jobs/api-testing-jobs>

50,000+ Api Testing jobs in United States (3,100 new) - L

50,000+ **Api Testing Jobs** in United States (3,100 new). Software Test Engineer. Engineer. HCL America, Inc. Menlo Park, CA \$85,000 - \$90,000.

<https://in.linkedin.com/jobs/api-testing-jobs>

17,000+ Api Testing jobs in India (437 new) - LinkedIn

17,000+ **Api Testing Jobs** in India (437 new) · Software Testing Engineer III · Software Quality Engineer - Remote · #2494 - Manual Functional Testing -Test Lead.

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Jobs

● Near Delhi, India

Past 3 days

Full-time

Work from home

No degree

Tester

Automation

QA- Automation Lead (API Testing+.java+Selenium)

<https://wingify.hire.trakstar.com/jobs/fk0udde/>



Salary of Testing

API Tester Salaries in India

Updated 14 Mar 2022

 Low Confidence

₹3,26,464/yr

Not enough reports to show salary distribution

Average Base Pay

1 salary

₹3L
Low

₹3L
High

No additional cash compensation has been reported for this role.

How much does an API Tester make? **The national average salary for an API Tester is ₹3,26,464 in India.** Filter by location to see API Tester salaries in your area. Salary estimates are based on 1 salaries submitted anonymously to Glassdoor by API Tester employees.

How accurate does ₹3,26,464 look to you?

 Right

 High

 Low

Your input helps Glassdoor refine our pay estimates over time.

What is the salary trajectory

https://www.glassdoor.co.in/Salaries/api-tester-salary-SRCH_KO0,10.htm#:~:text=The%20national%20average%20salary%20for,%E2%82%B93%2C26%2C464%20in%20India.



LIVE Research, Now See This

Click here

<https://www.linkedin.com/jobs/search/?currentJobId=3131328580&keywords=software%20tester>



Salary of Testing

Api Testing average salary in India 2022

Find out what the average Api Testing salary is

Type in a job title:

api testing

Find salary

Salary rate

Annual

Month

Biweekly

Weekly

Day

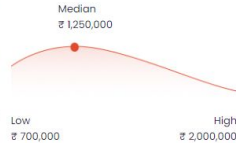
Hour

How much does a Api testing make in India?

₹ 1,250,000 / Annual

Based on 98 salaries

The average **api testing** salary in India is ₹ 1,250,000 per year or ₹ 641 per hour. Entry-level positions start at ₹ 700,000 per year, while most experienced workers make up to ₹ 2,000,000 per year.



<https://in.talent.com/salary?job=api+testing>

Do you know What is API?

Yes

No

In Comments

State of API Report 2023



1. APIs are rapidly growing.
2. New Tools are Coming to Test APIs
3. API Testers are needed or mandatory to learn now days.
4. API Automation is also important.
5. People are migrating test cases from, UI to APIs.

<https://sdet.live/api-report>



Why API Testing?

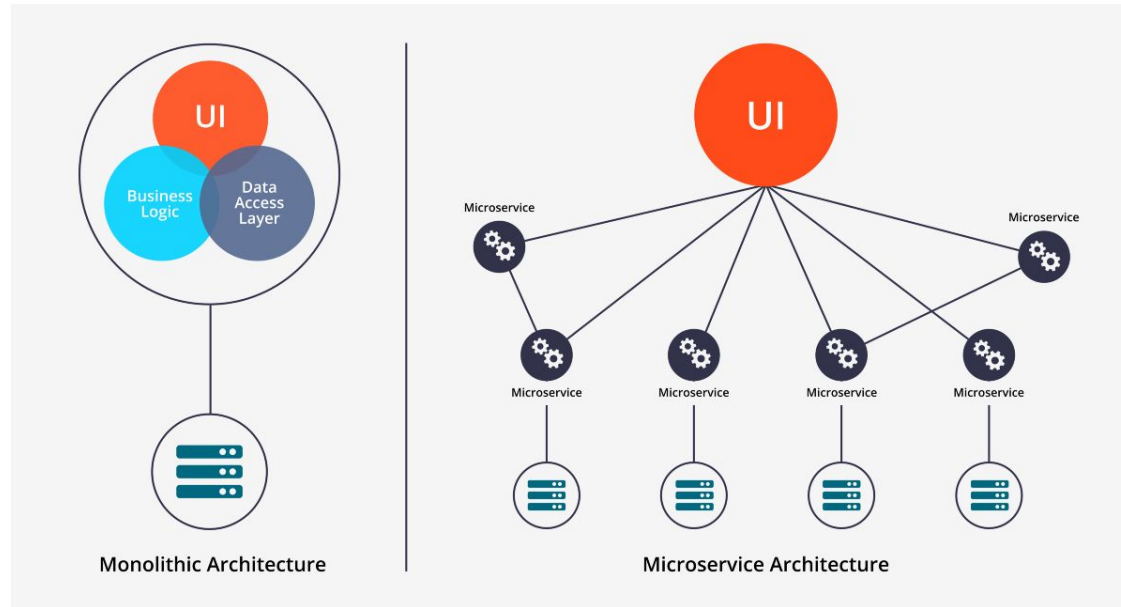
- Millions of Web Apps and Mobile Applications use APIs.





Why API Testing?

- MicroServices Architecture.





Microservices

Architectural style that structures an application as a collection of services

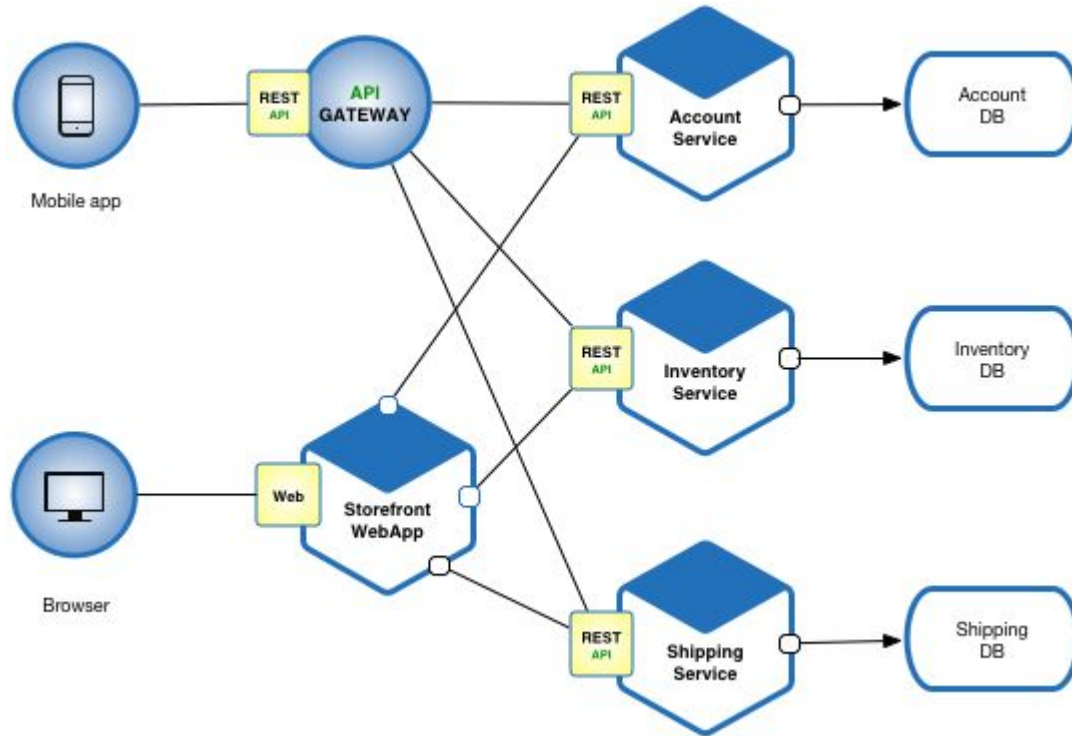
- Highly maintainable and testable
- Loosely coupled
- Independently deployable
- Organized around business capabilities
- Owned by a small team

Read more - <https://microservices.io/>





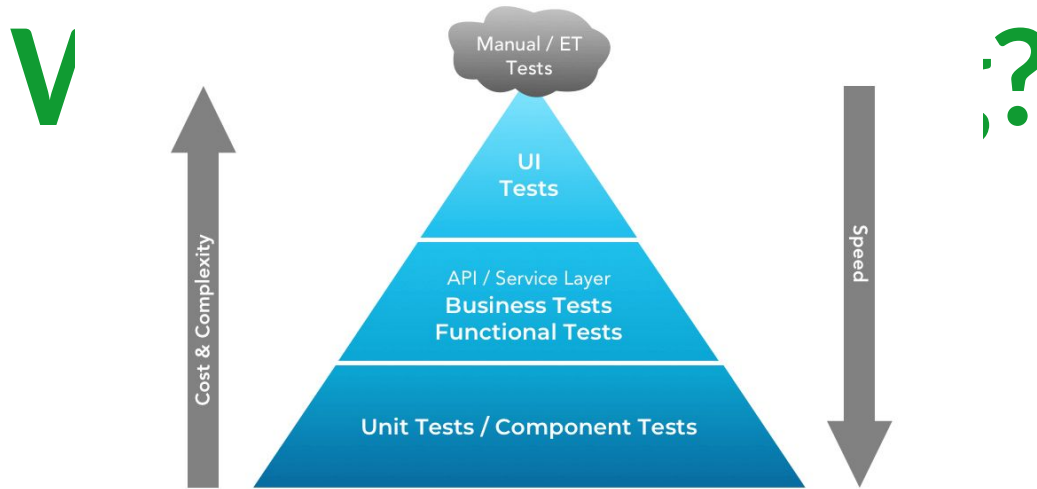
Microservices





Why API Testing?

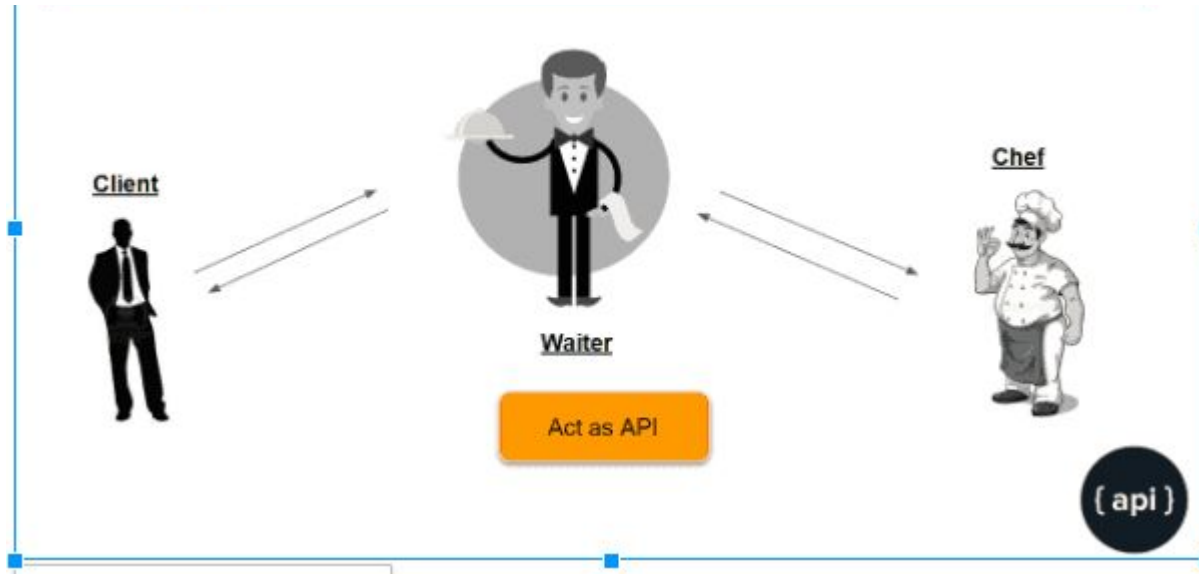
- UI Testing is costly.
- API Testing is easy and can caught early Bugs.





What is API ?

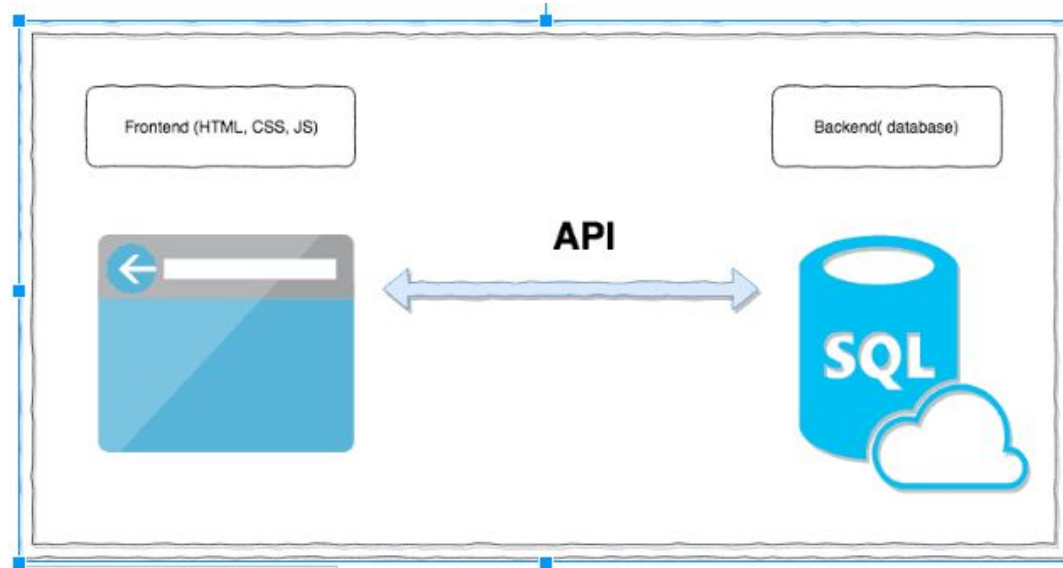
- API stands for the **Application Programming Interface**,





What is API ?

- Basically **a collection of** Functions and procedures which allows us to communicate two applications or libraries.



What are differences between API Testing and Unit Testing?



API Testing	Unit Testing
• Conducted by QA team	• Conducted by the development
• Mostly black box testing	• White box testing
• Aimed to assess the full functionality of the system for it will be employed by the end-user (external developers who will use your API)	• Used to verify whether each unit in isolation performs as expected or not
• Often run after the build is ready and authors do not have access to the source code	• Each of the code modules must be ensured to pass the unit test before being built by developers.

API Vs Web Service



- All Web Services are API, But All APIs are not web services.
- Web Services uses Three Styles,
- REST
- XML-RPC
- SOAP



- We are going to Learn REST API Testing.

Type of API



We are only Concern about the Web API

Simple Object Access Protocol

Remote Procedure Call

Representational State Transfer (REST).

etc.....



Web API

SOAP

RPC

REST

{ api }



SOAP



SOAP

- SOAP is an acronym for Simple Object Access Protocol.
- It is an XML-based messaging protocol for exchanging information among computers.
- SOAP is an application of the XML specification.



SOAP

- SOAP is a communication protocol designed to communicate via Internet.
- SOAP can extend HTTP for XML messaging.
- SOAP provides **data transport for Web services**.
- SOAP can exchange complete documents or call a remote procedure.
- SOAP can be used for broadcasting a message.
- SOAP is platform- and language-independent.
- SOAP is the XML way of defining what information is sent and how.
- SOAP enables client applications to easily connect to remote services and invoke remote methods



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SOAP

POST



https://www.dataaccess.com/webservicesserver/NumberConversion.wso

Params

Authorization

Headers (10)

Body ●

Pre-request Script

Tests ●

Settings



none



form-data



x-www-form-urlencoded



raw



binary



GraphQL

XML



```
1 <?xml version="1.0" encoding="utf-8"?>
2 <soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
3   <soap:Body>
4     <NumberToWords xmlns="http://www.dataaccess.com/webservicesserver/">
5       <ubiNum>502</ubiNum>
6     </NumberToWords>
7   </soap:Body>
8 </soap:Envelope>
```

<https://documenter.getpostman.com/view/611814/2s93Y5PeWV>



REST API?



What is REST API?

- REST is an acronym for REpresentational State Transfer.
- The REST architectural style describes six constraints.
- These constraints, put on the architecture, were initially communicated by Roy Fielding in his doctoral dissertation and defines the basis of RESTful-style.



What is REST API?

- In Simple Terms REST API is nothing but a Design Pattern for Web API



REST Constraints

- Uniform Interface
- Stateless
- Cacheable
- Client-Server
- Layered System
- Code on Demand (OPTIONAL)

<https://scrolltest.com/rest/explained>



Stateless

Stateless: It means that the necessary state to handle the request is contained within the request itself and server would not store anything related to the session.

In REST, the client must include all information for the server to fulfill the request whether as a part of query params, headers or URI.

Statelessness enables greater availability since the **server does not have to maintain, update or communicate that session state**



Cacheable

Every response should include whether the response is cacheable or not and for how much duration responses can be cached at the client side.

Client will return the data from its cache for any subsequent request and there would be no need to send the request again to the server.

But sometime there are chances that user may receive stale data.

Well-managed caching partially or completely eliminates some client-server interactions, further improving scalability and performance.



REST Used by

- Most of WebApps, Mobiles are relying on REST API

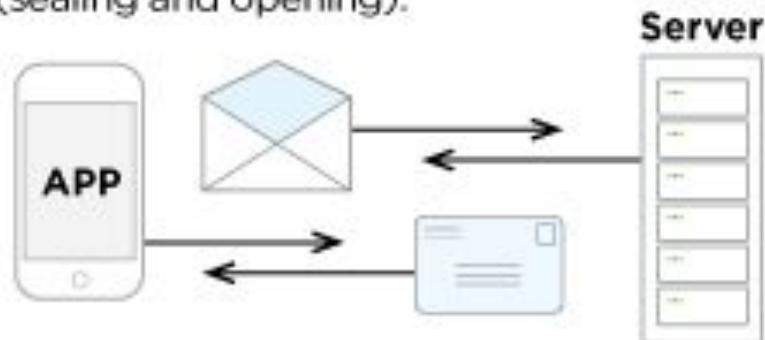




SOAP vs. REST APIs

SOAP is like using an envelope

Extra overhead, more bandwidth required, more work on both ends (sealing and opening).



REST is like a postcard

Lighterweight, can be cached, easier to update.



SOAP Vs REST

#	SOAP	REST
1	A XML-based message protocol	An architectural style protocol
2	Uses WSDL for communication between consumer and provider	Uses XML or JSON to send and receive data
3	Invokes services by calling RPC method	Simply calls services via URL path
4	Does not return human readable result	Result is readable which is just plain XML or JSON
5	Transfer is over HTTP. Also uses other protocols such as SMTP, FTP, etc.	Transfer is over HTTP only
6	JavaScript can call SOAP, but it is difficult to implement	Easy to call from JavaScript
7	Performance is not great compared to REST	Performance is much better compared to SOAP - less CPU intensive, leaner code etc.



What To Test in APIs?



What To Test in APIs?

- Verify the Parses JSON or XML response.
- HTTP Status Code.
- Verify Headers response with Negative or invalid headers.
- Max/ Min Key values.
- Have a test case to do XML, JSON Schema validation.
- Verify that how the APIs error codes handled.
- Missing Keys
- Empty Updates or POST call handling.
- Chaining request verification.
- End to End Crud.
- Database integrity test
- File upload.



What is Test in APIs?

- Test nulls and empty strings, these have a habitat of causing issues ;).
- Test with Unicode characters, ensure they are correctly saved to the database.
- Test mandatory and optional parameters, in particular test with just mandatory values.
- Ensure correct status codes are returned, you will find false positives e.g a 500 being returned as a 200.
- Functionality Bugs: the test looks for missing functionality bugs
- Reliability Bugs: API testing helps identify bugs pertaining to integration across different systems
- Performance Bugs: The tests help determine how much traffic the system can handle before being overloaded, and how to expand infrastructure to meet rising demand at the core, but also are very effective at pinpointing weak points in the API
- Security Bugs



Common Bugs in API?

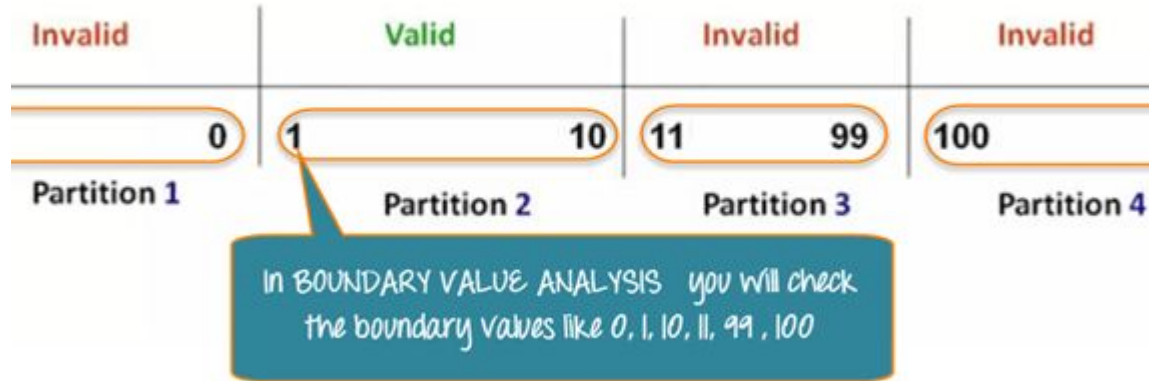
- Missing Keys
- Duplicate Keys
- Incorrect HTTP Code.
- HTTP Methods not handled properly
- Performance issues
- Security issues
- Validation errors
- API Monitoring issues

Manual Testing Concepts to Design API Test cases.

- Boundary Value Analysis (BVA)
- Equivalence Class Partitioning
- Decision Table based testing
- Error Guessing
- State Transition

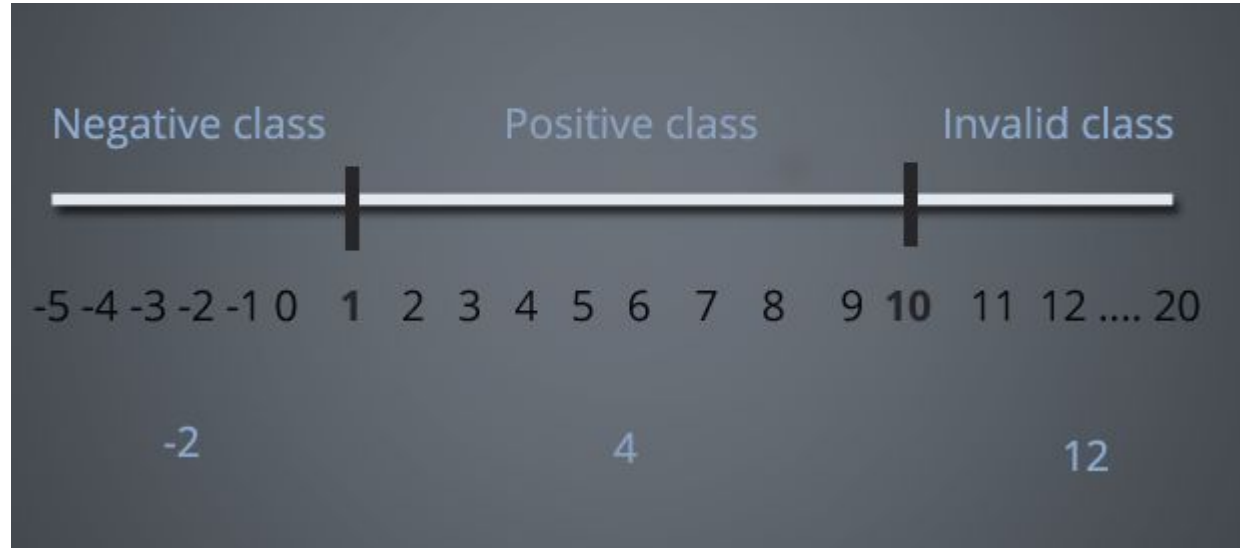


Boundary Value Analysis.





Equivalence Class Partitioning



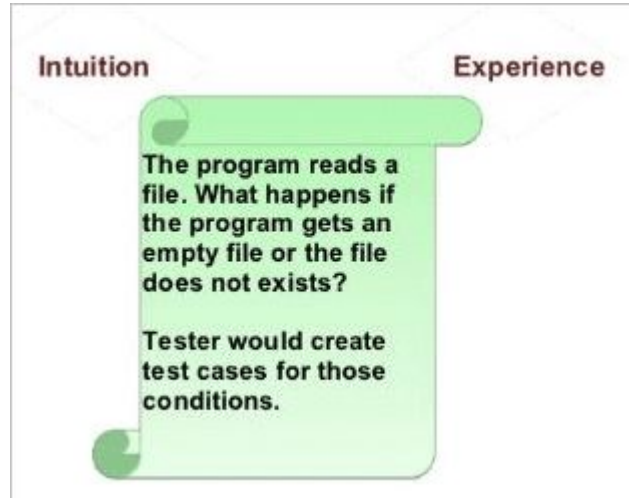


Decision Table based testing

ID	CONDITIONS/ACTIONS	TEST CASE 1	TEST CASE 2	TEST CASE 3	TEST CASE 4
Condition 1	Valid User ID	T	T	F	F
Condition 2	Valid Password	T	F	T	F
Action 1	Home Page	Execute			
Action 2	Show a Message as 'Invalid User Credentials'		Execute	Execute	Execute



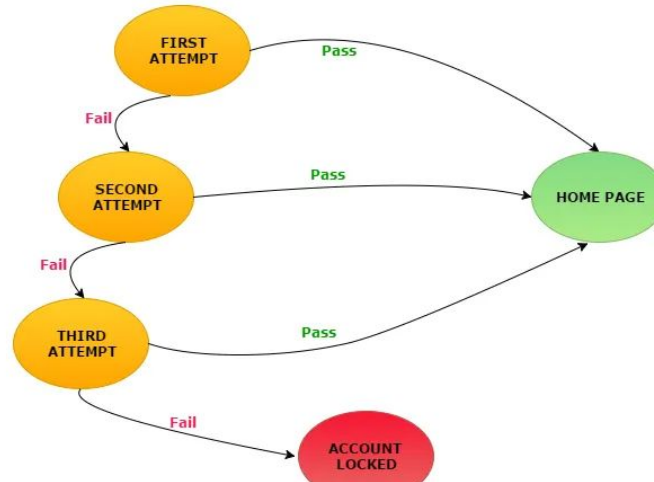
Error Guessing

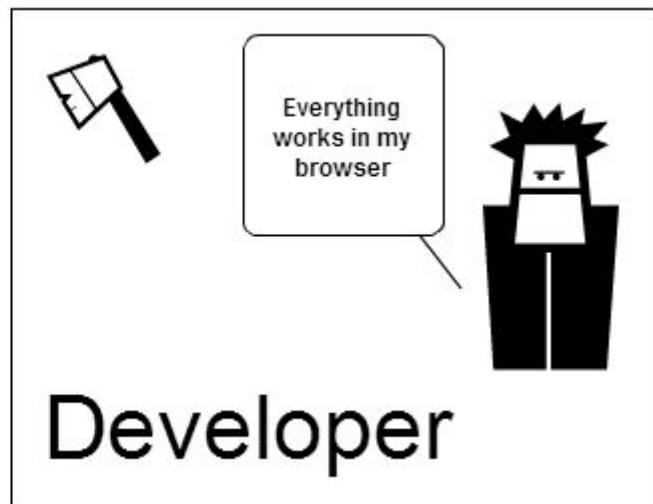




State Transition

Take an example of login page of an application which locks the user name after three wrong attempts of password.







Type of Testing in API Testing?



Functional Testing - POSTMAN

Integration Testing - POSTMAN

Regression Testing - POSTMAN

Security Testing - Burp Suite

Load Testing - JMeter

Penetration Testing - JMeter

Fuzz testing - injects invalid, malformed, or unexpected inputs into a system to reveal software defects



POSTMAN Basics



POSTMAN Basics

- OverView

- Import External Request.
- History
- WorkSpace in Postman
- Postman Tabs - Param, Auth, Headers, Body, Pre-req, Post Req(Test), Settings
- Postman Console.



Quiz

Can we use POST request to do all Request?

Thanks, for attending Class

**I hope you liked it. Say Thanks
in Comment :)**

Fin.